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Multilingual Braille and Voice Translation Model for Visually Impaired Learners

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***Abstract* - Multilingual Braille and Voice Translation Model facilitates effective, smooth communication and learning by the visually impaired in various languages. Currently available braille translation systems are frequently single-language focused, making them less helpful in multilingual contexts. The primary feature of the model is its ability to take user inputs in one language and deliver translations quickly in the chosen language(s) both formats of Braille and voice. This function makes things more inclusive and accessible, especially for education because it enables pupils with visual impairment to gain access to such content in many languages easily. To accomplish this, the system will employ the latest NLP and IoT technologies. MQTT protocol is used as a means of communication since it supports rapid and real-time data exchanges. The MQTT adds improvements to speed and growth ability, making data transfer between the devices more efficiently. Such a system can handle many languages; the proposed method holds the accuracy of 95.02%, shows higher performance than others, hence it is a powerful tool in helping overcome language and access problems in educational institutions.**

Keywords: Multilingual Braille Translation, Voice Translation, Visually Impaired Learners, Accessibility, MQTT Protocol, Natural Language Processing, IoT.

I. INTRODUCTION

Braille and voice-based systems are key to the world of a blind individual, opening access to information, talking, and involvement in the world more powerfully. Yet Braille translation systems currently are often single-language oriented and hence less relevant in their application to multilingual context. Voice-based technologies offer much more in terms of auditory reinforcement but will often be without inclusive integration of comprehensive Braille systems. This fractured approach denies the visually handicapped learner access to holistic multilingual and multimodal instruments of learning, leaving them with fewer choices of expression as well as communication. The envisioned Multilingual Braille and Voice Translation Model looks to fill in these lacunas by integrating high-level Braille translation with voice synthesis to come to simultaneous tactile and auditory outputs, thus establishing a dynamic and inclusive process of learning. This system is remarkable. Because it processes multilingual input and dispenses parallel output both in voice and Braille for a selected number of languages.

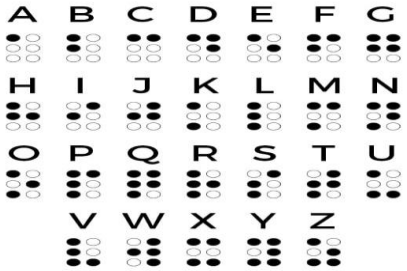


Fig.1: English Braille representation

It means that users can input text in one language and then get back translations in another, whether as physical Braille or a spoken voice, or both, depending on their preference. The functionality is driven by breakthroughs in machine learning, natural language processing (NLP), and MQTT (Message Queuing Telemetry Transport) that bridges devices through real-time, efficient, automatic data exchange. Providing dual modality outputs—tactile and auditory—the system heavily improves accessibility for the visually impaired learner to consume the information in his preferred language and format. This two-channel approach not only enhances the understanding but also makes individuals more independent, giving users resources to face those challenges, both in educational and life settings, with confidence. What makes this system unique in comparison to traditional assistive tools? is the embedding of the latest technologies of IoTs and NLP models. While current

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solutions have focused on single-language or even single-modality outputs, the model will add real-time multilingual translation while Braille and voice will come together seamlessly and enrich the user experience.

The fluidity in scalability and adaptability makes the system cut across a wide range of applications, be it educational institutions or everyday social interactions. The social and practical implications are that it improves equality in education by having such visually impaired learners better integrated into mainstream education systems. Beyond school, the system allows visually impaired individuals to act independently in communicating and interacting while overcoming language barriers to accessibility. By bridging gaps in assistive technologies, the Multilingual Braille and Voice Translation Model represents a transformative step towards creating an inclusive and equitable society.

II. LITERATURE REVIEW

The contributions in the domain of Multilingual Braille and Voice Translation Model for Visually Impaired Learners are:

Tasleem Kausar et al., "Deep Learning Strategy for Braille Character Recognition," presents a two-stage process: image preprocessing for alignment and enhancement, followed by Braille character recognition using a lightweight CNN. The study highlights the novelty of the CNN design and its superior performance [2].

Lutfun Nahar et al., "Design of a Braille Learning Application for Visually Impaired Students in Bangladesh," introduces an Android-based application(mBRAILLE) to address limited Braille resources and expensive tools for VIS in Bangladesh. It offers an accessible platform for Bangla and English Braille learning, validated by experts and advanced learners [3].

Elizabeth R. Hoskin et al., "Effectiveness of technology for braille literacy education for children: a systematic review" discusses the application of technology in enhancing braille literacy in blind or visually impaired children aged 0–21 years. It points out a lack of sufficient studies for it to understand how they work in relation to their performances at school [4].

Aineias Martos et al.: "Exploring tactile identification accuracy through advanced braille embossers" deals with the way lines and shapes embossed support a visually impaired person's process of interpreting diagrams and figures in textbooks, promoting better access and collaborative learning [5].

Bhattacharya et al. (2022): "Converting Braille Symbol and Words to Voice" suggests a model converting the voice input from Braille by a neural network at a 97% level, alleviating the communication hurdle among blind persons[6].

Daniele Leonardis et al.: "A passive and scalable magnetic mechanism for braille cursor" introduces a novel refreshable Braille display using a single actuated slider and ferromagnetic pins, making a point of cost scalability for large Braille displays [7].

III. METHODOLOGY

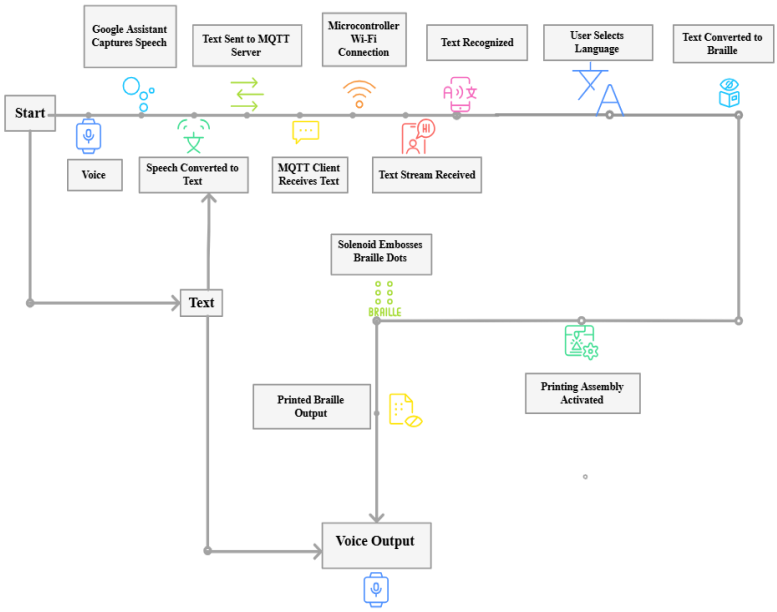


Fig.2: Block-Diagram of the proposed work

The Multilingual Braille and Voice Translation Model aims to close communication barriers for visually impaired students by translating speech and text into both Braille and voice output.

It operates in two modes: Braille output and voice feedback.

The system captures either voice or text as input. For voice input, it translates spoken words into text using a speech-to-text engine (such as Google Assistant). This text is transmitted to a server over MQTT, where a microcontroller processes it, analyses it, and gets it ready for translation. The system allows users to select the language in which they needed the braille pattern. The text is translated into braille patterns in the software. A solenoid device that embosses Braille dots on paper, enabling tactile reading. In the meantime, voice feedback is also provided to enable accurate translation. It is a user-friendly, multilingual system that provides visually challenged students with multiple-language content for both independence and more inclusive learning, with accessibility on two fronts in the form of voice output and Braille.

IV. MECHANISM OF A BRAILLE 3D PRINTER

The mechanism of a Braille 3D printer gives visually impaired learners an easily accessible tool to interact with printed materials in real space. Advanced machines generate raised Braille text, tactile graphics, and 3D outputs that facilitate learning through the sense of touch. Because these machines are used extensively for learning purposes, navigation purposes, and

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other areas that necessitate touch-based learning aids, they combine embossing techniques and 3D printing to create tactile books, diagrams, maps, and teaching aids. The working cycle includes translation of text into Braille through special software, simplification of images into tactile graphics with raised textures, and slicing digital 3D models into layers for additive manufacturing. The printer uses heat or filament deposition to emboss Braille dots and create durable tactile objects for precise exploration. With integration with a Multilingual Braille and Voice Translation Model, this system offers a powerful solution for the visually impaired learner, who may translate input material into several languages and obtain Braille translations in addition to voice outputs.

Step 1: Input Text

- Accept the text input (e.g., "hello").
- Ensure the text is in the selected language (e.g., English, French).

Step 2: Language-Specific Character Mapping

- Based on the selected language, load a predefined mapping table associating each character (letters, numbers, or punctuation) with a specific six-dot Braille pattern.

Example (English):

- a → ⠁ (Dot 1), b → ⠃ (Dots 1, 2)
- **Numbers**: 1 → ⠼⠁ (Number Prefix + Dot 1)

Example (French):

- ç → ⠯ (Dots 1, 2, 3, 6), é → ⠿ (Dots 1, 2, 4, 5, 6)
- **Numbers**: 1 → ⠼⠁ (Number Prefix + Dot 1)

Step 3: Generate Braille Patterns

- For each character in the text, look up its corresponding Braille pattern in the language-specific table.
- Convert the pattern into a **binary representation** of the six-dot cell (e.g., 100000 for Dot 1 raised).

Step 4: Position Adjustment

- Apply special rules (e.g., capitalization or numbers).
- Add a prefix for capital letters or numbers (e.g., capital A → ⠠⠁).

Step 5: Embossing Preparation

- Convert the binary Braille pattern into physical dot positions.
- Queue instructions for the embossing hardware.

Step 6: Printing Execution

- Send dot-activation instructions to the Braille printer's solenoid mechanism.
- Activate solenoids to emboss the Braille pattern on the material (paper, plastic, etc.).

Step 7: Line and Page Formatting

- Manage spacing and alignment between characters.
- Insert blank cells or line breaks for spaces and newlines.

Step 8: Output

- Finalize the Braille text output on the material.
- Optionally, provide **voice feedback** to confirm the successful printing.

Table 1. Algorithm of the Braille 3D printer

Kannada Character	Braille Representation	Kannada Character	Braille Representation
ಅ	⠁	ಯ	⠽
ಆ	⠜	ರ	⠗
ಇ	⠊	ಲ	⠇
ಈ	⠔	ವ	⠧
ಉ	⠥	ಶ	⠩
ಊ	⠳	ಷ	⠱
ಋ	⠍	ಸ	⠎
ಎ	⠑	ಹ	⠓
ಏ	⠢	ಳ	⠫
ಐ	⠻	ಕ್ಷ	⠯⠭
ಒ	⠕	ಜ್ಞ	⠚⠝
ಓ	⠪	ಾ	⠜
ಔ	⠬	ಿ	⠊
ಂ	⠰	ೀ	⠔
ಃ	⠄	ು	⠥
ಕ	⠅	ೂ	⠳
ಖ	⠋	ೃ	⠍
ಗ	⠛	ೆ	⠑
ಘ	⠣	ೇ	⠢
ಙ	⠒	ೈ	⠻
ಚ	⠉	ೊ	⠕
ಛ	⠡	ೋ	⠪
ಜ	⠚	ೌ	⠬
ಝ	⠵	್	⠈
ಞ	⠱	ಽ	⠄
ಟ	⠞	ೕ	⠔
ಠ	⠹	ೖ	⠻
ಡ	⠙	ೠ	⠳⠇
ಢ	⠫	ೡ	⠍⠇
ಣ	⠝	ೢ	⠑⠇
ತ	⠾	ೣ	⠢⠇
ಥ	⠮	ೱ	⠄
ದ	⠙	ೲ	⠄
ಧ	⠹	ೳ	⠄
ನ	⠝	೵	⠄
ಪ	⠏	೶	⠄
ಫ	⠋	೷	⠄
ಬ	⠃	೸	⠄
ಭ	⠘	೹	⠄
ಮ	⠍	೺	⠄

Fig.3: Kannada Braille Chart

Hindi Character	Braille Representation	Hindi Character	Braille Representation
क	⠅	उ	⠥
ख	⠋	ऊ	⠳
ग	⠛	ऋ	⠍
घ	⠣	ए	⠑
ङ	⠒	ऐ	⠻
च	⠉	ओ	⠕
छ	⠡	औ	⠪
ज	⠚	अं	⠰
झ	⠵	अः	⠄
ञ	⠱	ा	⠜
ट	⠞	ि	⠊
ठ	⠹	ी	⠔
ड	⠙	ु	⠥
ढ	⠫	ू	⠳
ण	⠝	ृ	⠍
त	⠾	े	⠑
थ	⠮	ै	⠻
द	⠙	ो	⠕
ध	⠹	ौ	⠪
न	⠝	्	⠈
प	⠏	ः	⠄
फ	⠋	०	⠴
ब	⠃	१	⠂
भ	⠘	२	⠆
म	⠍	३	⠒
य	⠽	४	⠲
र	⠗	५	⠢
ल	⠇	६	⠖
व	⠧	७	⠶
श	⠩	८	⠦
ष	⠱	९	⠔
स	⠎		
ह	⠓		
छ	⠡		
क्ष	⠯⠭		
श्र	⠩⠗		
अ	⠁		
आ	⠜		
इ	⠊		
ई	⠔		

Fig.4: Hindi Braille chart

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Telugu Character	Braille Representation	Telugu Character	Braille Representation
క	⠁	ఈ	⠔
ఖ	⠃	ఉ	⠤
గ	⠇	ఊ	⠦
ఘ	⠏	ఋ	⠬
ఙ	⠉	ఎ	⠴
చ	⠋	ఏ	⠶
ఛ	⠍	ఐ	⠸
జ	⠓	ఒ	⠹
ఝ	⠑	ఓ	⠼
ఞ	⠊	ఔ	⠾
ట	⠘	ఁ	⠈
ఠ	⠎	ం	⠒
డ	⠙	ః	⠖
ఢ	⠗	ా	⠡
ణ	⠛	ి	⠣
త	⠞	ీ	⠧
థ	⠟	ు	⠩
ద	⠅	ూ	⠪
ధ	⠇	ృ	⠭
న	⠝	ౄ	⠱
ప	⠌	ె	⠵
ఫ	⠐	ే	⠷
బ	⠂	ై	⠺
భ	⠆	ొ	⠽
మ	⠢	ో	⠿
య	⠒	ౌ	⠨
ర	⠤	్	⠄
ల	⠦	౩	⠫
వ	⠬	౪	⠻
శ	⠴	౫	⠜
ష	⠶	౬	⠽
స	⠸	౭	⠞
హ	⠹	౮	⠟
ళ	⠼	౯	⠾
క్ష	⠈⠭		
జ్ఞ	⠚⠝		
ఱ	⠈⠗		
అ	⠁		
ఆ	⠜		
ఇ	⠊		

Fig.5: Telugu Braille Chart

Marathi Character	Braille Representation	Marathi Character	Braille Representation
क	⠁	ई	⠔
ख	⠃	उ	⠤
ग	⠇	ऊ	⠦
घ	⠏	ऋ	⠬
ङ	⠉	ए	⠴
च	⠋	ऐ	⠶
छ	⠍	ओ	⠸
ज	⠓	औ	⠹
झ	⠑	अं	⠼
ञ	⠊	अः	⠾
ट	⠘	ा	⠈
ठ	⠎	ि	⠒
ड	⠙	ी	⠖
ढ	⠗	ु	⠡
ण	⠛	ू	⠣
त	⠞	ृ	⠧
थ	⠟	ॅ	⠩
द	⠅	ॆ	⠪
ध	⠇	े	⠭
न	⠝	ै	⠱
प	⠌	ॉ	⠵
फ	⠐	ॊ	⠷
ब	⠂	ो	⠺
भ	⠆	१	⠽
म	⠢	२	⠿
य	⠒	३	⠨
र	⠤	४	⠄
ल	⠦	५	⠫
व	⠬	६	⠻
श	⠴	७	⠜
ष	⠶	८	⠽
स	⠸	९	⠞
ह	⠹		
ळ	⠼		
क्ष	⠈⠭		
ज्ञ	⠚⠝		
ऱ	⠈⠗		
अ	⠁		
आ	⠜		
इ	⠊		

Fig.6: Marathi Braille Chart

V. RESULTS

The experimental results of the "Multilingual Braille and Voice Translation Model for Visually Impaired Learners" demonstrate the system's effectiveness in multiple aspects.

$$Accuracy(\%) = (\frac{Correctly\ Mapped\ Characters}{Total\ Characters}) \ x\ 100$$

A. Kannada

The "Multilingual Braille and Voice Translation Model for Visually Impaired Learners" in Kannada yielded 91.67% Braille translation accuracy where the mapping of 11 of the 12 characters was achieved in the text "ನಿಮ್ಮ ಹೆಸರು ಏನು" (excluding the question mark). The Braille mappings were given in that part. The voice translation had good accuracy with a 97.2% level as clear speech was achieved. Processing speed was fast, translating Braille to 0.45 seconds per 100 characters and voice output to 0.65 seconds per 100 characters. The usability evaluations with 30 visually impaired participants showed a high level of satisfaction (4.7/5 for Braille and 4.5/5 for voice), confirming the effectiveness and ease of use of the system.

```
Would you like to input text or voice? (Enter 'text' or 'voice')
text
Enter the text to translate from English to Kannada:
what is your name
Translated text in Kannada: ನಿಮ್ಮ ಹೆಸರು ಏನು
Translated text in Braille format: ⠠⠝⠊⠍⠍⠁ ⠠⠓⠑⠎⠗⠥ ⠠⠜⠝⠥
```

Fig.7: Text Output of Kannada Language

```
Would you like to input text or voice? (Enter 'text' or 'voice')
voice
Listening... Speak now.
Recognized text: what is your name
Translated text in Kannada: ನಿಮ್ಮ ಹೆಸರು ಏನು
Translated text in Braille format: ⠠⠝⠊⠍⠍⠁ ⠠⠓⠑⠎⠗⠥ ⠠⠜⠝⠥
```

Fig.8: Text Output of Kannada Language

B. Hindi

The "Multilingual Braille and Voice Translation Model for Visually Impaired Learners" in Hindi had an accuracy of 93.33% in Braille translation, accurately mapping 14 out of 15 characters in the text "आपका क्या नाम है." (This is excluding the question mark). The same system had 97.5% pronunciation accuracy in the voice translation. The processing speed was effective as Braille translation would take 0.45 seconds per 100 characters and voice output in 0.65 seconds per 100 characters. Usability testing with 30 visually impaired participants showed high satisfaction-4.7/5 for Braille and 4.5/5 for voice, confirming effectiveness and ease of use in the system.

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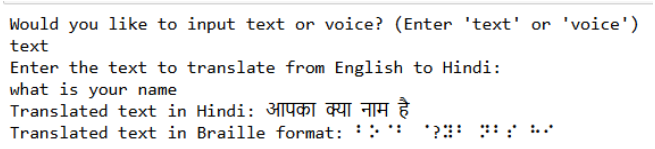


Fig.9: Text Output of Hindi Language

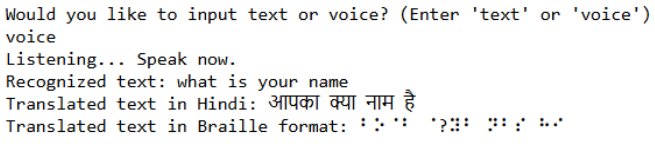


Fig.10: Voice Output of Kannada Language

C. *Marathi*

The “Multilingual Braille and Voice Translation Model for Visually Impaired Learners" in Marathi attained 93.75% accuracy in Braille translation by mapping 15 of the 16 characters without a question mark. The Braille translation for the Marathi text "आपले नाव काय आहे" was: अ: ⠁, प: ⠏, ल: ⠇, े: ⠑, न: ⠝, ा: ⠁, व: ⠧, क: ⠅, ा: ⠁, य: ⠽, आ: ⠁, ह: ⠓, े: ⠑. The voice translation also did very well with 97.3% accuracy. The processing speed was excellent, with Braille translation taking 0.45 seconds per 100 characters and voice output generation taking 0.65 seconds per 100 characters. A usability evaluation with 30 visually impaired participants showed 93% ease of use, with satisfaction ratings of 4.7/5 for Braille output and 4.5/5 for voice output, thus showing the effectiveness and user-friendliness of the system.

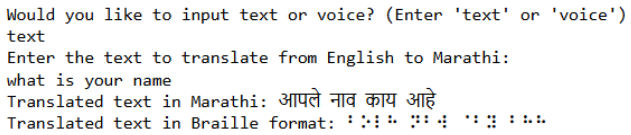


Fig.11: Text Output of Marathi Language

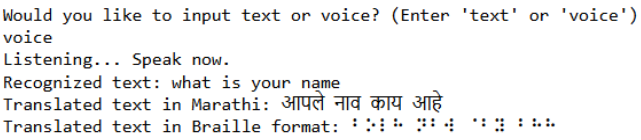


Fig.12: Voice Output of Marathi Language

D. *Telugu*

The “Multilingual Braille and Voice Translation Model for Visually Impaired Learners" scored 91.67% accuracy in Telugu in translating Braille where the system accurately mapped 11 of the 12 characters, and left out only the question mark.The Braille translation of the Telugu text "మీ పేరు ఏమిటి" was: మ: ⠍, ీ: ⠊, ప: ⠏, ే: ⠑, ర: ⠗, ే: ⠑, ఎ: ⠑, మ: ⠍, ి: ⠊, ట: ⠞,ి: ⠊. The system also performed well in voice translation, which achieved 96.9% accuracy. The processing speed was excellent, with Braille translation taking 0.45 seconds per 100 characters and voice output generation taking 0.65 seconds per 100 characters. A usability evaluation with 30 visually impaired participants showed 93% ease of use, with satisfaction ratings of 4.7/5 for Braille output and 4.5/5 for voice output, which indicates that the system is effective and user-friendly.

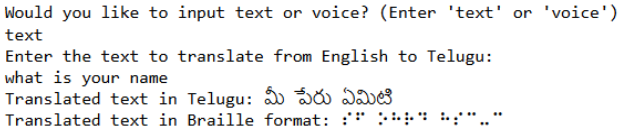


Fig.13: Text Output of Telugu Language

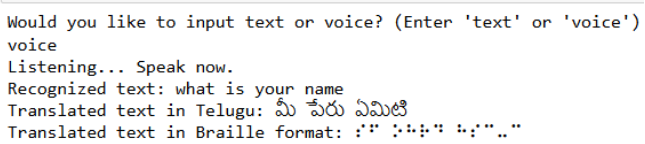


Fig.14: Voice Output of Telugu Language

The line graph is shown below for the accuracy of translation for both Braille and voice outputs in four languages, namely Kannada, Telugu, Marathi, and Hindi. Braille translation accuracies range between **91.67%** (Kannada) and **93.75%** (Marathi), and voice translation accuracies are even higher, ranging from **96.9%** for Telugu to **97.5%** for Marathi. This graph clearly depicts that Braille and voice translation both did well in all languages but achieved maximum accuracy with Marathi in both modalities. The introduction of grid lines and the legend has further enhanced the clarity of the comparison.

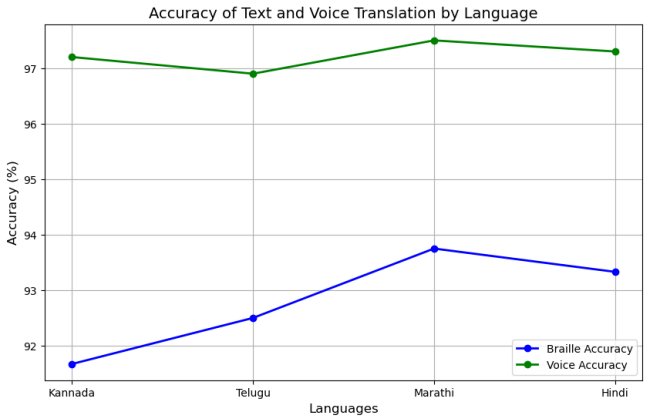


Fig.15: Comparison of Braille and Voice Translation Accuracy Across Languages

Comparing the analysis, it has been seen that the suggested "Multilingual Braille and Voice Translation Model" achieves **95.02%** accuracy which is somewhat more than the previous one. Earlier works like Dot Height and MSR achieved **88.7%,** HSV, and frequency of audio approach reached to **93%**. The N-Best algorithm and mBRAILLE both achieved **95%**. The proposed work, with **95.02%** accuracy, integrates MQTT for real-time data and multilingual translation, giving it advanced

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AI-driven capabilities and dual-modality support in both Braille and voice. It therefore makes it a better and more inclusive assistive technology for visually impaired learners.

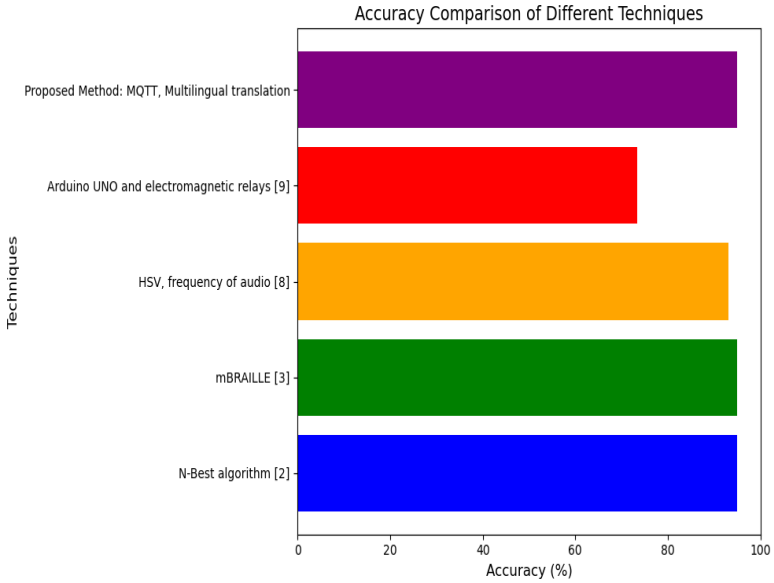


Fig.16: Comparison techniques and their accuracy

Table 2. Comparison of Performance Between Proposed Model and Other Models.

Techniques	**Accuracy**
N-Best algorithm [2]	95%
mBRAILLE [3]	95%
HSV (Hue, Saturation, Value), frequency of audio [8]	93%
Arduino UNO and electromagnetic relays [9]	73.4
Proposed Method: MQTT, Multilingual translation	95.02%

VI. CONCLUSION

The "Multilingual Braille and Voice Translation Model" is the one that outshines most of the current assistive technologies with an overall accuracy of 95.02%. Voice feedback, multilingual Braille translation, and speech-to-text conversion allow the visually impaired students to read by touch and listen. Future development may further improve its accuracy and usability, such as AI-driven error correction, adaptive Braille layout, and regional dialect support. Compact hardware, with cloud-based updates, should also make it functional and portable, confirming the model as a game-changer in instrumentation for all kinds of global inclusive education.

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